

Product Requirements Document (PRD)

Product Name

FoundryERP

A modern Manufacturing ERP and Inventory Management System for Steel Casting Foundries.

1. Product Vision

FoundryERP is a lightweight, modern, web-based manufacturing ERP designed specifically for steel foundries and casting manufacturers.

The primary goal is to help foundry owners track:

- Raw Materials
- Bill of Materials (BOM)
- Material Consumption
- Production Orders
- Finished Goods
- Inventory Valuation

while providing a simple and intuitive dashboard for owners and managers.

This is NOT intended to be a generic ERP like SAP or Odoo.

The system is optimized for manufacturing operations where raw materials are consumed to produce finished products.

2. Problem Statement

Most foundries currently use:

- Excel Sheets
- Paper Registers
- WhatsApp
- Tally

As a result, management struggles to answer questions such as:

- How much steel scrap do we currently have?

- How much material was consumed this month?
- Which products consume the most raw material?
- What is the cost of producing a product?
- Which production orders are currently running?
- What inventory is available right now?

The software should provide these answers instantly.

3. Core Business Flow

Raw Material Purchase



Inventory Updated



Create BOM



Create Production Order



Material Consumption



Finished Goods Produced



Inventory Updated



Reports & Analytics

4. User Roles

Admin

Full system access.

Can manage:

- Users
 - Roles
 - System Settings
 - Dynamic Fields
 - BOM Configuration
-

Owner

Can view:

- Dashboard
- Reports
- Inventory
- Production
- Orders

Cannot modify system settings.

Manager

Can manage:

- Inventory
 - Production Orders
 - Purchases
 - Customers
-

Operator

Can:

- Start Production
- Complete Production

- Record Production Data
-

5. Functional Requirements

Module 1: Authentication

Features

- Login
- Logout
- JWT Authentication
- Protected Routes
- Role Based Access Control

Pages

/login

/register

Module 2: Dashboard

Purpose

Provide business visibility within 10 seconds.

KPI Widgets

- Total Inventory Value
- Raw Material Value
- Finished Goods Value
- Production Today
- Production This Month
- Pending Orders
- Low Stock Alerts
- Top Consumed Materials

Charts

- Monthly Material Consumption
- Monthly Production
- Inventory Value Trend

- Top Consumed Materials
 - Order Status Distribution
-

Module 3: Raw Material Inventory

Examples

- Steel Scrap
- Pig Iron
- Ferro Silicon
- Ferro Manganese

Fields

- Material Code
- Material Name
- Category
- Unit
- Current Stock
- Minimum Stock
- Average Cost
- Storage Location

Features

- Create Material
 - Edit Material
 - Delete Material
 - Search Materials
 - Filter Materials
 - Stock In
 - Stock Out
 - Inventory Ledger
 - Low Stock Alerts
-

Module 4: Product Master

Examples

- Valve Body
- Pump Housing
- Impeller

Fields

- Product Code
- Product Name
- Description
- Unit
- Standard Cost
- Selling Price

Features

- Create Product
 - Edit Product
 - Delete Product
-

Module 5: Bill of Materials (BOM)

Business Importance

This is the most critical module in the entire system.

Purpose

Define what raw materials are required to produce a product.

Example

Product:

Valve Body

Materials:

Steel Scrap = 20 KG

Ferro Silicon = 2 KG

Ferro Manganese = 1 KG

Output Quantity:

10 Units

Features

- Create BOM
 - Edit BOM
 - Delete BOM
 - Clone BOM
 - BOM Versioning
 - BOM Cost Calculation
 - Material Requirement Calculation
-

Dynamic BOM Column Engine

Requirement

Users must be able to add new BOM columns without developer involvement.

Example

Default BOM Columns

- Material
- Quantity Required
- Unit

Later User Wants

- Heat Number

User Action

Settings

→ BOM Configuration

→ Add Column

Field Name: Heat Number

Field Type: Text

Save

Updated BOM Columns

- Material

- Quantity Required
- Unit
- Heat Number

Supported Types

- Text
- Number
- Currency
- Date
- Boolean
- Dropdown
- Multi Select
- Textarea
- Attachment

Additional Configuration

- Required
- Unique
- Searchable
- Filterable
- Visible
- Editable

Nice To Have

Extend dynamic fields to:

- Products
- Inventory
- Production Orders
- Purchase Orders

Module 6: Production Orders

Workflow

Sales Order



Production Order



Material Consumption



Finished Goods



Inventory Updated

Fields

- Production Number
- Product
- BOM Version
- Quantity To Produce
- Status

Statuses

- Planned
- In Progress
- Completed
- Cancelled

Features

- Create Production Order
- Start Production
- Pause Production
- Complete Production

Module 7: Automatic Material Consumption Engine

Business Rule

When a production order is completed:

System should:

1. Read BOM
2. Calculate Material Requirements
3. Deduct Materials
4. Create Inventory Transactions
5. Create Audit Logs
6. Create Finished Goods

Example

BOM:

10 Valve Bodies

Requires

20 KG Steel Scrap

2 KG Ferro Silicon

1 KG Ferro Manganese

Production Order:

100 Valve Bodies

System Automatically Consumes:

200 KG Steel Scrap

20 KG Ferro Silicon

10 KG Ferro Manganese

Inventory Updated Automatically

Module 8: Finished Goods Inventory

Features

- Automatic Stock Increase
- Inventory Valuation
- Search
- Filters

Fields

- Product
 - Available Quantity
 - Reserved Quantity
 - Inventory Value
-

Module 9: Purchase Management

Features

- Purchase Orders
- Goods Receipt
- Supplier Management
- Purchase Returns

Workflow

Purchase Order

↓

Receive Materials

↓

Inventory Updated

Module 10: Sales Orders

Features

- Create Order
- Edit Order
- Order Tracking

Fields

- Customer
 - Product
 - Quantity
 - Due Date
 - Status
-

Module 11: Inventory Ledger

Purpose

Track every stock movement.

Transaction Types

- Purchase
- Consumption
- Production
- Adjustment
- Transfer
- Return

Store

- Reference Number
 - User
 - Timestamp
 - Before Quantity
 - After Quantity
-

Module 12: Reporting

Reports

- Inventory Report
 - Production Report
 - Consumption Report
 - Material Usage Report
 - Stock Valuation Report
 - BOM Cost Report
 - Purchase Report
 - Sales Report
-

Module 13: Audit Logs

Track

- Create
- Update
- Delete

Store

- User
 - Timestamp
 - Old Value
 - New Value
-

6. Non Functional Requirements

Performance

- Dashboard < 2 seconds
- Inventory Search < 500ms
- Production Order Creation < 1 second

Security

- JWT Authentication
- Password Hashing
- Role Based Permissions

Scalability

Support:

- 100+ Concurrent Users
- 1 Million+ Inventory Records

Availability

99.9% uptime target

7. Technology Stack

Frontend

- React
- TypeScript
- Tailwind CSS
- Zustand
- React Query
- Recharts

Backend

- Node.js
- Express
- TypeScript

Database

- MongoDB Atlas

Authentication

- JWT
-

8. MVP Scope

Must Have

- ✓ Inventory
- ✓ Product Master
- ✓ BOM
- ✓ Dynamic BOM Columns
- ✓ Production Orders
- ✓ Material Consumption Engine
- ✓ Finished Goods Inventory
- ✓ Dashboard
- ✓ Reports

Nice To Have

- Dynamic Fields Across All Modules
- Workflow Builder
- Notification Center
- Mobile App

The MVP success criterion is that a foundry owner can accurately track how much raw material is consumed to produce finished goods and view inventory levels in real time.